

Linear regression

The mathematical tool behind this model is the **conditional expectation**:

$$E(X|Y=y) : R^k \rightarrow R , y = (y_1, \dots, y_k)'$$

$$E(X|Y=y) = \int_R x \cdot \frac{f(x, y_1, \dots, y_k)}{f(y_1, \dots, y_k)} dx ,$$

where the marginal density of Y is $f(y_1, \dots, y_k) = \int_R f(x, y_1, \dots, y_k) dx$.

The function $y \rightarrow E(X|Y=y)$ is called the regression of X in Y .

Linear regression

The model of linear regression is the following:

$$E(X|Y=y) = \beta_0 + \beta_1 y_1 + \cdots + \beta_k y_k,$$

where $\beta_0, \beta_1, \dots, \beta_k$ are the regression (real) parameters.

called *intercept*

The regression hyperplane has the equation:

$$x = \beta_0 + \beta_1 y_1 + \cdots + \beta_k y_k.$$

Suppose that the statistical data are:

$$(x_i, y_{i1}, \dots, y_{ik}), \quad i = 1, \dots, n \quad n > k.$$

Linear regression

Under the assumption of linearity, the regression parameters can be determined by the Least Squares method:

$$SS(\beta_0, \beta_1, \dots, \beta_k) = \sum_{i=1}^n (x_i - \beta_0 - \beta_1 y_{i1} - \dots - \beta_k y_{ik})^2$$

$$\frac{\partial SS}{\partial \beta_0} = \frac{\partial SS}{\partial \beta_1} = \dots = \frac{\partial SS}{\partial \beta_k} = 0 \Leftrightarrow$$

Linear regression

$$\left\{ \begin{array}{l} \sum_{i=1}^n (x_i - \beta_0 - \beta_1 y_{i1} - \cdots - \beta_k y_{ik}) = 0 \\ \sum_{i=1}^n y_{ij} (x_i - \beta_0 - \beta_1 y_{i1} - \cdots - \beta_k y_{ik}) = 0, \quad j = 1, \dots, k \end{array} \right. \Rightarrow$$

$\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ the *LS* estimators of the regression parameters

(the solution is unique if $\text{rank} \left(\begin{array}{c} y_{ij} \\ i=1, \dots, n \\ j=1, \dots, k \end{array} \right) = k < n$).

Important obs. This hypothesis of linearity cannot be intuitively identified by visual inspection of the data for $k \geq 2$.

In the case of multivariate normal distribution of the data, the hypothesis of linearity holds.

Linear regression

Proposition. Suppose that $(X, Y_1, \dots, Y_k)'$ is a normally distributed random vector of dimension $k+1$:

$$(X, Y_1, \dots, Y_k)' \sim N(\mu, \Sigma),$$

where $\mu = (\mu_x, \mu'_y)'$ and $\Sigma = \begin{pmatrix} \sigma_x^2 & \Sigma_{x,y} \\ \Sigma'_{x,y} & \Sigma_y \end{pmatrix}$, with $\Sigma_y = \text{cov}(Y, Y)$,

$$\Sigma_{x,y} = \text{Cov}(X, Y'), \quad Y = (Y_1, \dots, Y_k)'$$

Then, the conditional expectation $E(X|Y)$ is linear:

$$E(X|Y=y) = \Sigma_{x,y} \Sigma_y^{-1} y + (\mu_x - \Sigma_{x,y} \Sigma_y^{-1} \mu_y).$$

The regression hyperplane of X in Y can be written as:

$$x - \mu_x = \Sigma_{x,y} \Sigma_y^{-1} (y - \mu_y).$$

(for details see Dumitrescu & Bătătorescu, p. 219-221)

Linear regression

Obs. For $k=1$, $(X, Y)'$ is bidimensional.

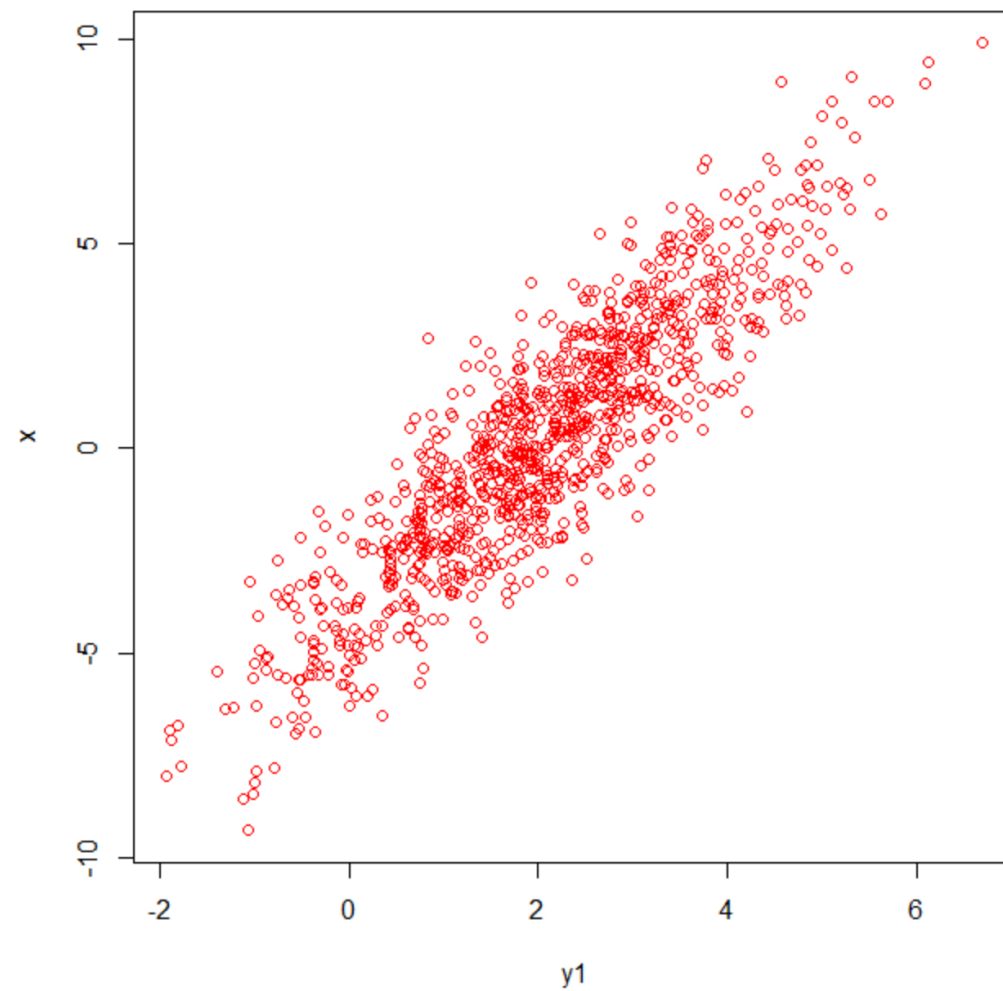
$(X, Y)' \sim N(\mu, \Sigma)$, where $\mu = (\mu_x, \mu_y)'$, $\Sigma = \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix}$ and $\rho = \frac{\sigma_{xy}}{\sigma_x\sigma_y}$ is the correlation coefficient.

The conditional expectation becomes $E(X|Y=y) = \mu_x + \rho \frac{\sigma_x}{\sigma_y} (y - \mu_y)$ and the regression line can be written as:

$$x - \mu_x = \rho \frac{\sigma_x}{\sigma_y} (y - \mu_y).$$

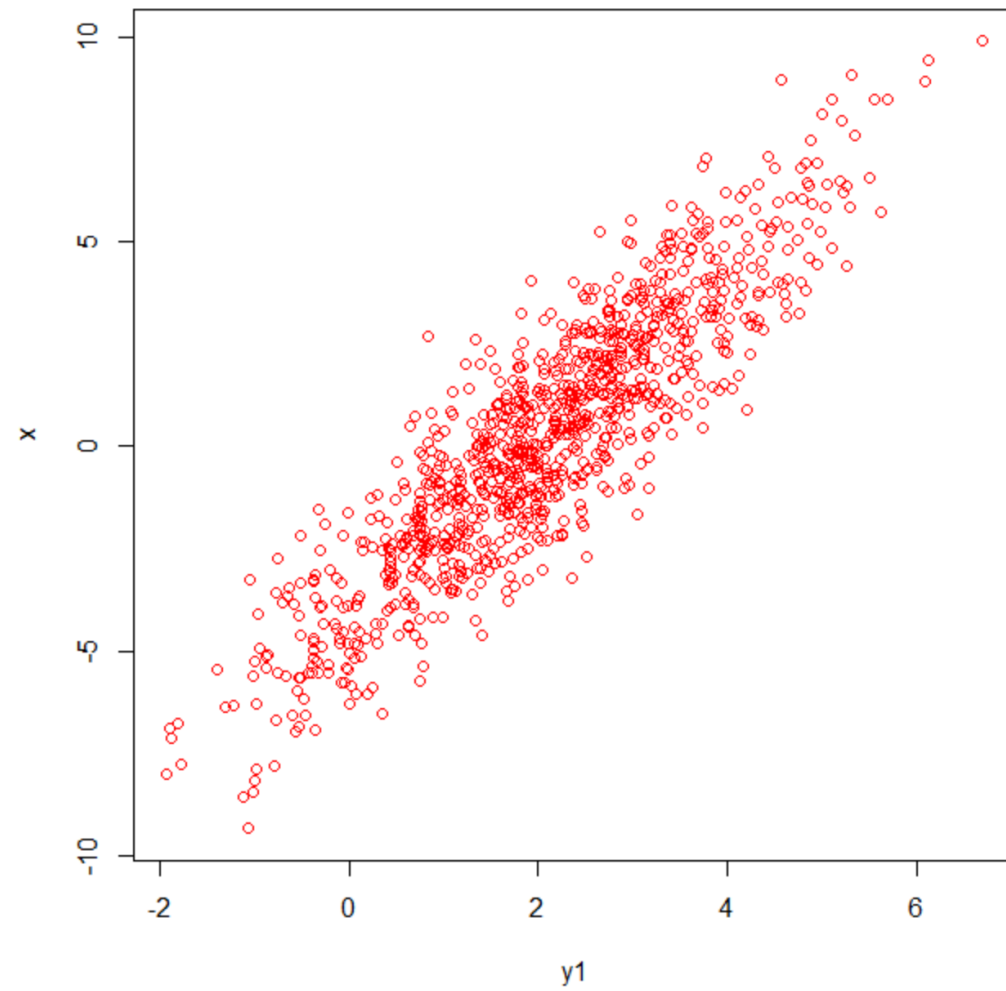
Linear regression

```
library(MASS)
mu=c(0,2)
Sigma=matrix(c(10,4,4,2),2,2)
X<-mvrnorm(1000,mu,Sigma)
x<-X[,1]
y1<-X[,2]
plot(y1,x,col="red")
```

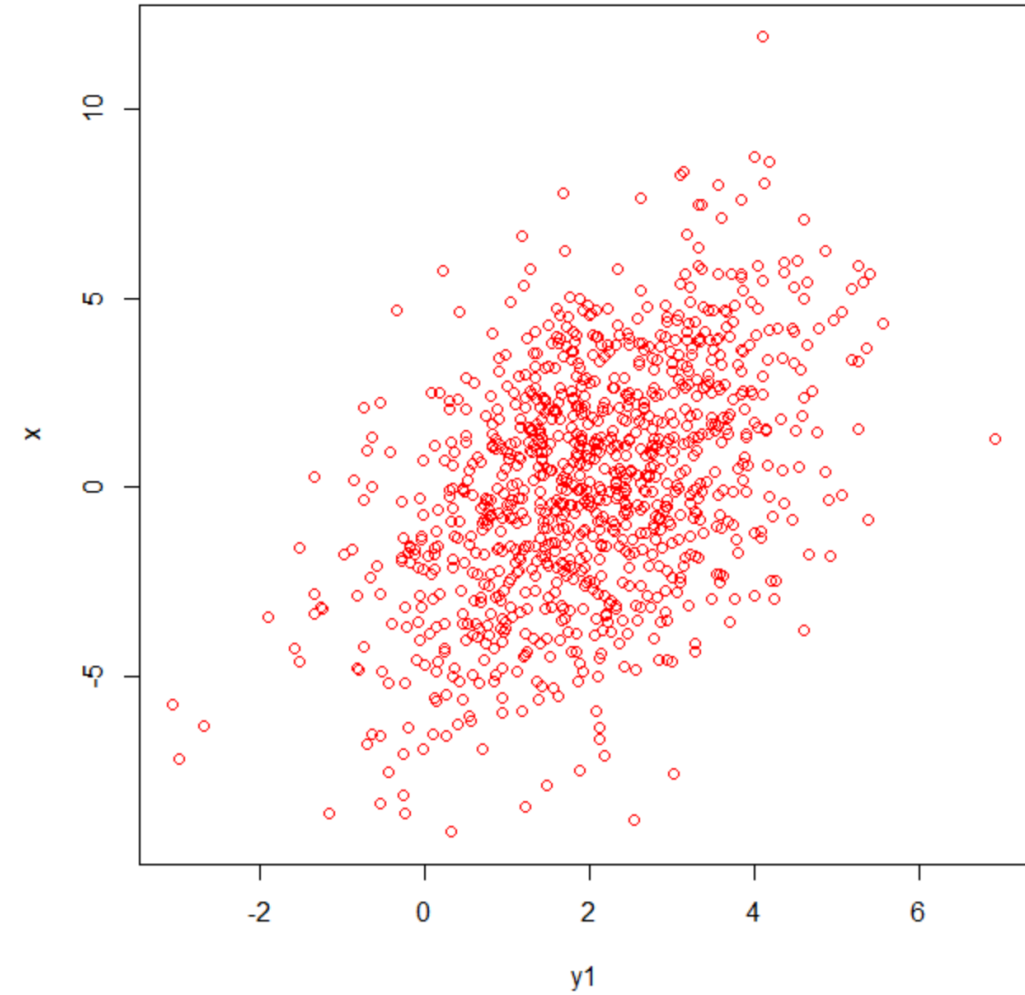


Linear regression

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Linear regression

Statistical inference for the regression line

For the bidimensional random vector $(X, Y)'$, we consider the linear regression model

$$E(X|Y=y) = \beta_0 + \beta_1 y,$$

with the regression line

$$x = \beta_0 + \beta_1 y.$$

Suppose that $(X_i, Y_i)'_{i=1, \dots, n}$ are observations *i.i.d.* $\sim (X, Y)'$ and $(x_i, y_i)'_{i=1, \dots, n}$ are the corresponding statistical data.

Linear regression

For statistical data, we consider the linear models

$$X_i = \beta_0 + \beta_1 y_i + \xi_i, \quad i = 1, \dots, n,$$

where $\xi_1, \dots, \xi_n$ are *i.i.d.* with the conditional mean 0 and the conditional variance σ^2 .

We use the Least Square method to estimate the parameters β_0 and β_1 .

$$SS(\beta_0, \beta_1) = \sum_{i=1}^n (x_i - \beta_0 - \beta_1 y_i)^2.$$

$$\frac{\partial SS}{\partial \beta_0} = \frac{\partial SS}{\partial \beta_1} = 0 \Leftrightarrow \begin{cases} n\beta_0 + \beta_1 \sum_{i=1}^n y_i = \sum_{i=1}^n x_i \\ \beta_0 \sum_{i=1}^n y_i + \beta_1 \sum_{i=1}^n y_i^2 = \sum_{i=1}^n x_i y_i \end{cases}$$

The system has a unique solution $(\hat{\beta}_0, \hat{\beta}_1)$, assuming that $\text{not}(y_i = \bar{y}, \forall i)$.

Linear regression

Notations:

$$L_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i$$

$$L_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i - \bar{x})x_i$$

$$L_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (y_i - \bar{y})y_i$$

The unique solution is:

$$\hat{\beta}_1 = \frac{L_{xy}}{L_{yy}}, \quad \hat{\beta}_0 = \bar{x} - \hat{\beta}_1 \bar{y},$$

and the regression line becomes:

$$x - \bar{x} = \frac{L_{xy}}{L_{yy}} (y - \bar{y}).$$

Linear regression

The LS estimators of the parameters of the regression line are:

$$\hat{\beta}_1(X_1, \dots, X_n) = \frac{\sum_{i=1}^n (X_i - \bar{X})(y_i - \bar{y})}{L_{yy}}$$

$$\hat{\beta}_0(X_1, \dots, X_n) = \bar{X} - \hat{\beta}_1(X_1, \dots, X_n)\bar{y}$$

The estimators have the following properties (in terms of the conditional distribution):

$$E(\hat{\beta}_1) = \beta_1, \quad E(\hat{\beta}_0) = \beta_0,$$

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{L_{yy}}, \quad Var(\hat{\beta}_0) = \frac{\sigma^2}{n} + \frac{\sigma^2 \bar{y}^2}{L_{yy}},$$

$$Cov(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{y}}{L_{yy}}.$$

Linear regression

Now we perform the analysis of variance for the linear regression ($k=1$).

The fitted (or predicted) values for the regression line are defined as:

$$\hat{X}_i = \hat{\beta}_0 + \hat{\beta}_1 y_i, \quad i = 1, \dots, n$$

We define the following sums of squares:

$$SS_t = \sum_{i=1}^n (X_i - \bar{X})^2$$

$$SS_{regression} = \sum_{i=1}^n (\hat{X}_i - \bar{X})^2 - \text{how far the slope of the regression line is from 0.}$$

$$SS_{residual} = \sum_{i=1}^n (X_i - \hat{X}_i)^2 - \text{how close the sample points are to the regression line.}$$

Linear regression

For a good-fitting model, we want a high $SS_{regression}$ and a low $SS_{residual}$.

$$R^2 \stackrel{\text{def}}{=} \frac{SS_{regression}}{SS_t}$$

if the model fits perfect then $R^2 = 1$.

if $R^2 = 0$ it means that y gives no information about x .

Obs. R^2 increases by adding new features in the model (even if the feature added is pure noise!). The solution – use it together with adjusted R^2 that penalizes for the number of feature added $R^2_{adj} = 1 - [(1 - R^2)(n - 1)/(n - k - 1)]$.

Situation	R^2	Adjusted R^2
Add a useful feature	Increases	Increases
Add a useless feature	Increases slightly	Decreases or stays same
Add many useless features	Keeps increasing	Drops significantly

table generated by Claude

Linear regression

By direct calculation, we get that:

$$SS_t = L_{XX}, \quad SS_{regression} = \frac{L_{Xy}^2}{L_{yy}}, \quad SS_{residual} = L_{XX} - \frac{L_{Xy}^2}{L_{yy}}.$$

(hence $SS_t = SS_{regression} + SS_{residual}$).

$$\text{Notation: } \overline{SS} = \frac{1}{\text{deg_of_freedom}} \cdot \frac{SS}{\sigma^2}$$

The hypothesis $H_0: \{\beta_1 = 0\}$ can be tested by using the F -test.

Proposition. Under the assumption of normality for $(X, Y)'$ and if the hypothesis $H_0: \{\beta_1 = 0\}$ is true, then

$$\frac{\overline{SS}_{regression}}{\overline{SS}_{residual}} = \frac{\frac{1}{1} \cdot \frac{1}{\sigma^2} \cdot SS_{regression}}{\frac{1}{n-2} \cdot \frac{1}{\sigma^2} \cdot SS_{residual}} \sim F_{1, n-2}$$

(for details see Dumitrescu & Bătătorescu, p. 223-224).

Linear regression

The test statistic for

$$H_0: \{\beta_1 = 0\} \text{ against } H_a: \{\beta_1 \neq 0\}$$

is $\frac{\overline{SS}_{regression}}{\overline{SS}_{residual}}$, and the acceptance region is

$$W_{n;1-\varepsilon} = \left\{ (x_1, y_1, \dots, x_n, y_n) \mid \frac{\overline{SS}_{regression}}{\overline{SS}_{residual}} \leq f_{1,n-2;1-\varepsilon} \right\},$$

where ε is the significance level and $f_{1,n-2;1-\varepsilon}$ is the $(1 - \varepsilon)$ quantile of the $F_{1,n-2}$ distribution.

The hypothesis $H_0: \{\beta_1 = 0\}$ against $H_a: \{\beta_1 \neq 0\}$ can also be tested by a t -test.

Linear regression

Proposition. Under the assumption of normality for $(X, Y)'$ and if the hypothesis $H_0: \{\beta_1 = 0\}$ is true, then:

$$\tilde{t} \stackrel{\text{not}}{=} \frac{\hat{\beta}_1}{\sqrt{\frac{1}{n-2} \cdot SS_{\text{residual}} \cdot \frac{1}{L_{yy}}}} \sim t_{n-2}.$$

The acceptance region of the t -test at the significance level ε is:

$$W'_{n;1-\varepsilon} = \left\{ (x_1, y_1, \dots, x_n, y_n) \mid -t_{n-2;1-\frac{\varepsilon}{2}} \leq \tilde{t} \leq t_{n-2;1-\frac{\varepsilon}{2}} \right\},$$

where $t_{n-2;1-\frac{\varepsilon}{2}}$ is the superior $\frac{\varepsilon}{2}$ quantile of the t_{n-2} distribution.

Linear regression

Interval estimation for β_0, β_1

Proposition. Under the assumption of normality for $(X, Y)'$, the following random variables are distributed t_{n-2} :

$$\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{1}{n-2} \cdot SS_{residual} \cdot \frac{1}{L_{yy}}}} \sim t_{n-2}$$

$$\frac{\hat{\beta}_0 - \beta_0}{\sqrt{\frac{1}{n-2} \cdot SS_{residual} \cdot \left(\frac{1}{n} + \frac{\bar{y}^2}{L_{yy}} \right)}} \sim t_{n-2}$$

Linear regression

For the confidence level $1 - \varepsilon$, the confidence intervals are:

$$C_{n;1-\varepsilon}^{(\beta_1)} = \left\{ -t_{n-2;1-\frac{\varepsilon}{2}} \leq \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{1}{n-2} SS_{residual} \frac{1}{L_{yy}}}} \leq t_{n-2;1-\frac{\varepsilon}{2}} \right\}$$
$$= \left\{ \hat{\beta}_1 - t_{n-2;1-\frac{\varepsilon}{2}} \sqrt{\frac{1}{n-2} SS_{residual} \frac{1}{L_{yy}}} \leq \beta_1 \leq \hat{\beta}_1 + t_{n-2;1-\frac{\varepsilon}{2}} \sqrt{\frac{1}{n-2} SS_{residual} \frac{1}{L_{yy}}} \right\}.$$

$$C_{n;1-\varepsilon}^{(\beta_0)} = \left\{ -t_{n-2;1-\frac{\varepsilon}{2}} \leq \frac{\hat{\beta}_0 - \beta_0}{\sqrt{\frac{1}{n-2} SS_{residual} \left(\frac{1}{n} + \frac{\bar{y}^2}{L_{yy}} \right)}} \leq t_{n-2;1-\frac{\varepsilon}{2}} \right\}$$
$$= \{ \dots \leq \beta_0 \leq \dots \}.$$

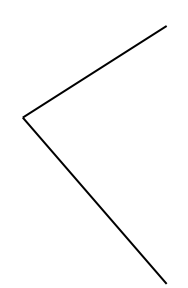
Linear regression

The sample correlation coefficient $r = \frac{L_{xy}}{\sqrt{L_{xx} L_{yy}}}$.

The relationship between r and $\hat{\beta}_1$ is :

$$\hat{\beta}_1 = r \frac{\sqrt{L_{xx}}}{\sqrt{L_{yy}}}$$

We have that $-1 \leq r \leq 1$

 $r > 0$ ($r < 0$) then X tends to increase (resp. decrease) as Y increases.

$r = 0$ then X is unrelated to Y .

If we are interested in whether there is an association between two variables, then we analyze r .

If we are interested in prediction, then we take a look at $\hat{\beta}_1$.

Linear regression

Summary (for $k=1$)

- we have a set of data $\{(x_1, y_1), \dots, (x_n, y_n)\}$, $x_i, y_i \in \mathbb{R}$
- we fit a linear regression model to our data - we estimate the parameters β_0 and β_1 .
- if we consider that $SS_{residual}$ is “small enough”, then we make the assumption that the data are normally distributed.

Linear regression

Summary (for $k=1$)

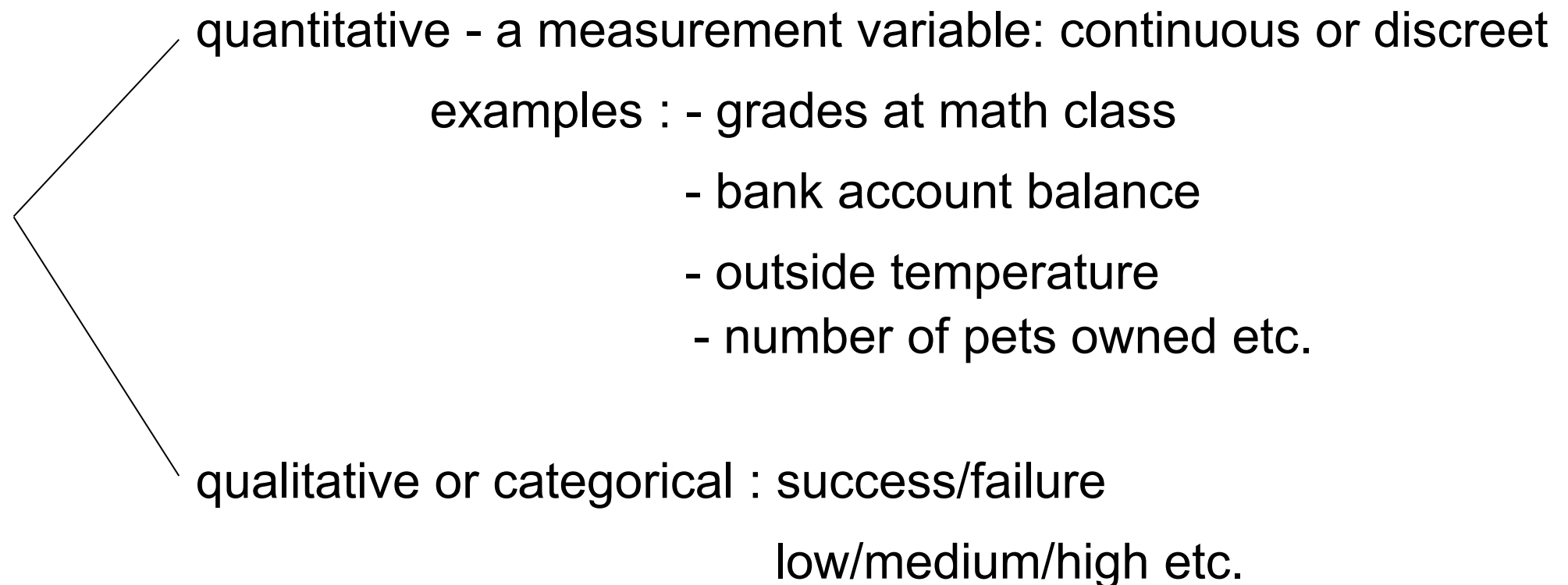
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- we fit a linear regression model to our data - we estimate the parameters β_0 and β_1 .
- if we consider that $SS_{residual}$ is “small enough”, then we make the assumption that the data are normally distributed.
- under the assumption of normality, we perform ANOVA.

This analysis is important because if $\beta_1 = 0$ then our data are uncorrelated – either the linearity hypothesis does not hold or the data are independent – in both cases, we will not apply a linear regression model for our data.

Linear regression

Comment

There are two types of variables used in statistics :



Regression = model for the relationship “causes” → “effects”.

Linear regression

Depending on the type of the causes and effects, we have different situations that are differently treated:

- 1) The effect is qualitative; the causes are quantitative
-the typical model used in the Generalized Linear Regression
(e.g., Logistic Regression, Log-Linear Regression).
- 2) The effect is quantitative continuous; the causes are quantitative - the typical model is Linear Regression.